

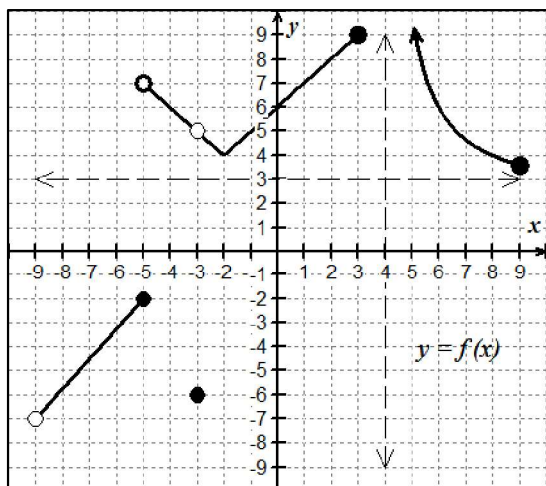
Lesson 6: Limits and Continuity

Now that we have an understanding of limits and using limit notation, we can use limits to define continuity at a point. ***This will be a very important concept throughout our course.***

Topic 1.11: Defining Continuity at a Point

EX #1: A Discovery Exploration.

Use the graph below to complete the table. You should ***look for three conditions that are necessary to satisfy the definition of continuity. That is, what three conditions must exist in order for $f(x)$ to be continuous at a point $x = c$?***



Definition of Continuity

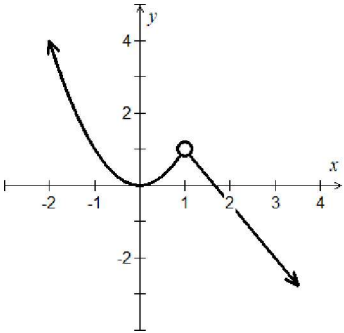
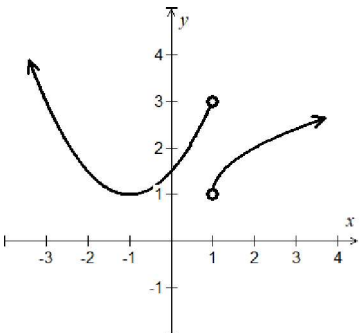
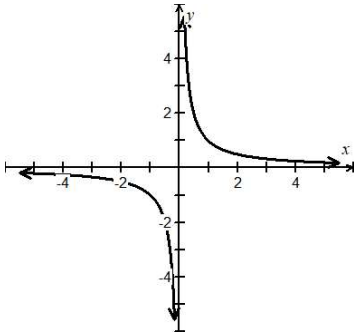
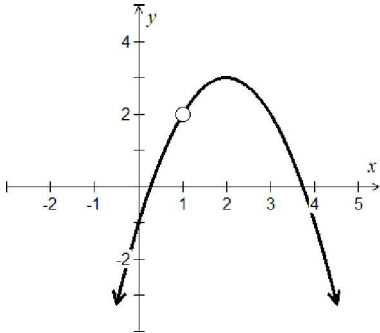
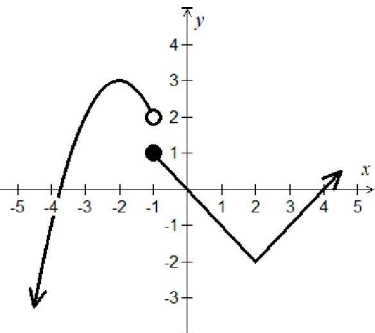
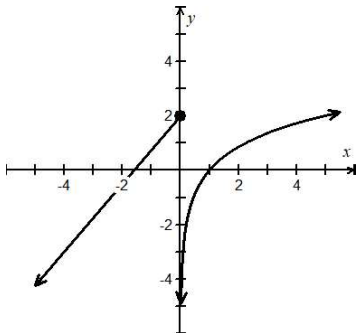
$x = c$	Find the function value, if it exists, for given x-value	Find left-hand limit $\lim_{x \rightarrow c^-} F(x)$	Find right-hand limit $\lim_{x \rightarrow c^+} F(x)$	Find general limit $\lim_{x \rightarrow c} F(x)$	Is $F(x)$ continuous at $x = c$?
$x = 9$					
$x = 4$					
$x = 3$					
$x = 0$					
$x = -3$					
$x = -5$					

**Analyze this table and write the three conditions for continuity above.
Memorize the rules!**

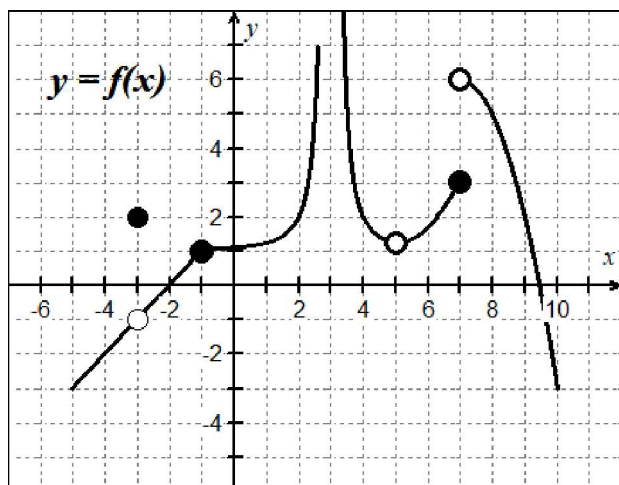
While **limits** told us where a function intended to go.
Continuity guarantees that the function actually made it there.

Classifying Discontinuities

Topic 1.10: Exploring Types of Discontinuities

Removable or Point (Holes) 2-sided limit exists	Non-Removable	
	Jump 1-sided limits exists	Infinite At least one of the 1-sided limits doesn't exist
		
		

EX #2: Find the points (intervals) at which the function is continuous, and the points at which the function is discontinuous on the interval $-5 < x < 10$.



Definition of Continuity - More Facts and Theorems

One-Sided Continuity

A function $f(x)$ is called

- ☐ Left-continuous at $x = c$ if $\lim_{x \rightarrow c^-} f(x) = f(c)$
- ☐ Right-continuous at $x = c$ if $\lim_{x \rightarrow c^+} f(x) = f(c)$

Continuity at a Point

Suppose $f(x)$ is defined on an open interval containing $x = c$.

Then f is continuous at $x = c$ if

$$\lim_{x \rightarrow c} f(x) = f(c)$$

Continuity on an Open Interval

A function is continuous on an open interval (a, b) if it is continuous at each point in the interval.

Continuity on a Closed Interval

A function is continuous on a closed interval $[a, b]$ if it is continuous on the open interval (a, b) and the function is continuous from the right at a and continuous from the left at b .

Continuity Laws of Some Basic Functions

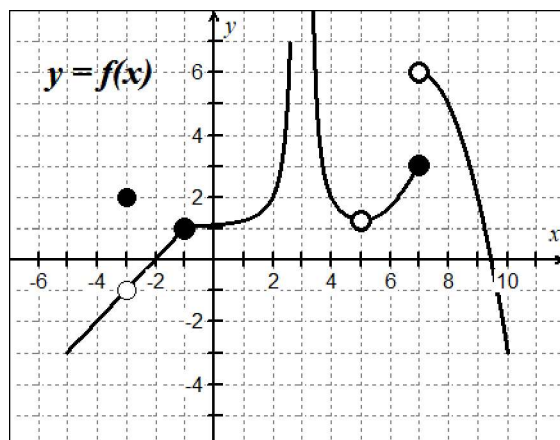
- ☐ Polynomial functions $P(x)$ are continuous over reals
- ☐ Rational Functions $P(x)/Q(x)$ is continuous on its domain such that $Q(x) \neq 0$.
- ☐ $y = x^{1/n}$ is continuous on all reals if n is odd and continuous on $[0, \infty)$ if n is even.
- ☐ $y = \sin x$ and $y = \cos x$ are continuous over reals
- ☐ $y = b^x$ is continuous for $b > 0, b \neq 1$
- ☐ $y = \log_b x$ is continuous for $x > 0, b > 0, b \neq 1$
- ☐ Inverse functions - if $f(x)$ is continuous on an interval with range R and $f^{-1}(x)$ exists, then $f^{-1} \circ f$ is continuous on domain R .

Properties of Continuity

Given functions f and g continuous at $x = c$, then the following functions are also continuous at $x = c$.

1. Scalar multiple: $b \cdot f$
2. Sum or difference: $f \pm g$
3. Product: $f \cdot g$
4. Quotient: $\frac{f}{g}$; if $g(c) \neq 0$
5. Compositions: If g is continuous at c and f is continuous at $g(c)$, then the composite function is continuous at c , $(f \circ g)(x) = f(g(x))$

EX #3: For $c = -3$, $c = 3$, and $c = 7$, find $f(c)$, $\lim_{x \rightarrow c^-} f(x)$, $\lim_{x \rightarrow c^+} f(x)$, and $\lim_{x \rightarrow c} f(x)$. Justify your findings using the three-part definition of continuity.



Topic 1.12: Confirming Continuity over an Interval

You must be able to confirm continuity without a graph or a calculator by using your knowledge of function behavior for parent functions and their transformations.

EX #4: Find the values of x where the function is discontinuous. Describe the type, infinite or removable. Justify your answer by using the definition of continuity.

A. $f(x) = \frac{1}{x-3}$

B. $g(x) = \frac{2x^2 + 7x + 6}{x+2}$

Topic 1.13: Removing Discontinuities

EX #4: Use the definition of continuity to find the value of k so that the function is continuous for all real numbers.

A. $g(x) = \begin{cases} kx^2, & x \leq 2 \\ kx - 6, & x > 2 \end{cases}$

B. $h(x) = \begin{cases} \frac{|x - 4|}{x - 4}, & x < 4 \\ 5k - 4x, & x \geq 4 \end{cases}$

EX #5: Given $h(x) = \begin{cases} -2x - 5 ; & x < -2 \\ 3 & ; & x = -2 \\ x^3 - 6x + 3; & x > -2 \end{cases}$ for what values of x is $h(x)$ not continuous? Justify.

EX #6: Use the three-part definition of continuity to create a system of equations. Then, find the values of a and b so that $f(x)$ is continuous for all real numbers.

$$f(x) = \begin{cases} 2ax - b & ; \quad x < -1 \\ 6 & ; \quad x = -1 \\ ax^2 + b & ; \quad x > -1 \end{cases}$$

You know that the function value and the limit value can exist independently of each other. Let's summarize the big ideas of continuity on intervals, at points, and one-sided limits.

EX #7: Given $g(x) = \begin{cases} 2x + 1, & x < 3 \\ x^2, & x \geq 3 \end{cases}$ find each of the following.

A. $\lim_{x \rightarrow 3^-} g(x)$

B. $\lim_{x \rightarrow 3^+} g(x)$

C. $g(3)$

D. $\lim_{x \rightarrow 3} g(x)$

F. Is $g(x)$ continuous at $x = 3$? Justify.

Summary:

If a function is continuous at a point, then the function value and the limit value are the same at that point!